E2.1.1 The photoelectric effect and threshold frequency

The photoelectric effect is emission of electrons when light falls on a surface. A photon's energy must reach the surface's work function for emission, giving a threshold frequency. Below threshold, increasing light intensity still produces no photoemission.

Once frequency reaches threshold, increasing intensity at fixed frequency increases the incident photon rate and usually the emitted electron rate. Emission can begin immediately because one absorbed photon can supply the energy needed for escape.

Photocurrent-voltage graphs, saturation and stopping potential

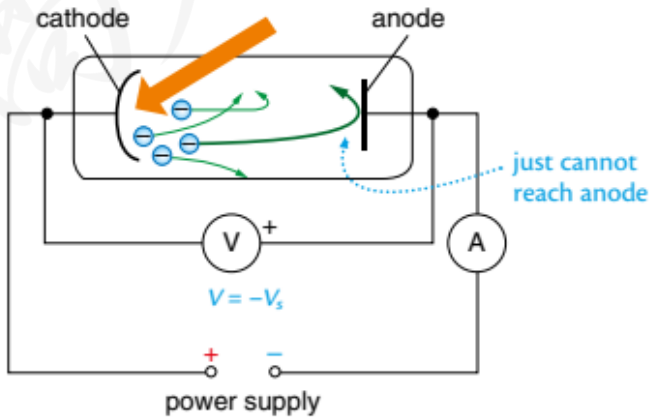
Let V be the collector potential relative to the emitter. For V > 0, the field accelerates photoelectrons and more reach the collector, increasing current. Once all emitted photoelectrons are collected, current reaches saturation and stays constant as voltage increases further.

For V < 0, the field slows photoelectrons. When the reverse-voltage magnitude reaches stopping potential V_s, even the most energetic electrons just fail to reach the collector and current is zero. $eV_s = K_{max}$ gives their maximum kinetic energy at emission.

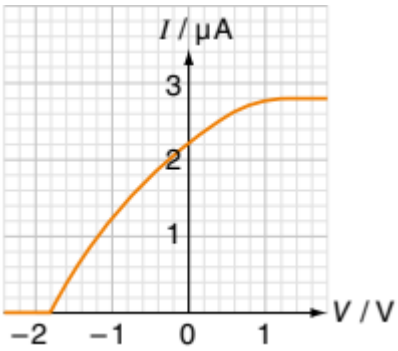
For V > 0, the maximum arrival kinetic energy is $K_{max} + eV$. Increasing intensity at fixed frequency raises saturation current, while emission energy and stopping potential stay unchanged.

$$K_{max} = eV_s$$

K_{max}: maximum photoelectron kinetic energy (J); V_s: stopping potential magnitude (V); e = 1.60 × 10^{-19} C, positive elementary charge.



Reverse voltage prevents photoelectrons reaching the collector; at the stopping potential the photocurrent just reaches zero.



The zero-current intercept gives stopping potential; the plateau gives saturation current.

E2.1.2 Photon energy

Light exchanges energy in photons. Each photon's energy is proportional to frequency and inversely proportional to vacuum wavelength. Higher frequency gives more energy per photon. Increasing intensity at fixed frequency mainly increases photon number.

At incident power P, the photon rate is $P/(hf)$. Raising frequency at fixed total power reduces the photon rate; photoelectron rate also depends on surface absorption and emission. With the SI Planck constant, use wavelength in metres to obtain joules, then convert to eV if needed.

$$E = hf = \frac{hc}{\lambda}$$

Photon energy E (J); h = 6.63 × 10^{-34} J s; f: Hz; c ≈ 3.00 × 10^{8} m s^{-1}; λ: vacuum wavelength (m).

E2.1.3 Work function and threshold frequency

Work function is the minimum energy needed to free an electron from a surface, depending on material and surface condition. A threshold-frequency photon just supplies that energy, so maximum emitted kinetic energy tends to zero. Keep work function, photon energy and kinetic energy consistently in joules or electronvolts.

$$\phi = hf_0$$

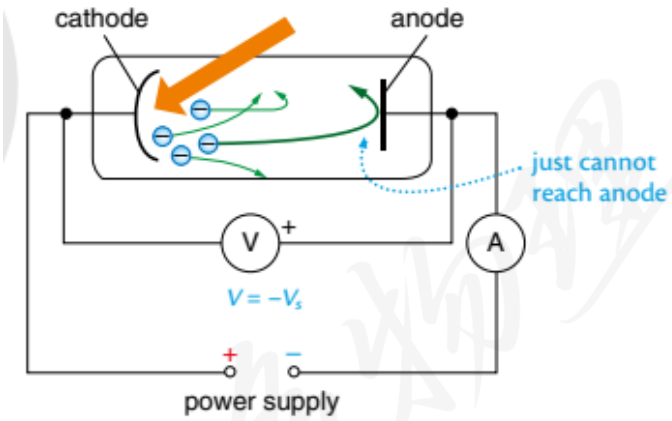
φ: work function (J); f_0: threshold frequency (Hz); h = 6.63 × 10^{-34} J s. Use consistent energy units.

E2.1.4 Einstein's photoelectric equation

A photon's energy first overcomes the work function. The remainder is the maximum photoelectron kinetic energy. Higher frequency increases this maximum, whereas higher intensity at fixed frequency does not. A negative calculated result means insufficient photon energy and no emission, not electrons with negative kinetic energy.

$$K_{\max} = hf - \phi = h(f - f_0)$$

Photoemission requires $f \geq f_0$. K_{max} and φ: J; f, f_0: Hz; h = 6.63×10^{-34} J s. A negative result means no photoemission, not negative kinetic energy.



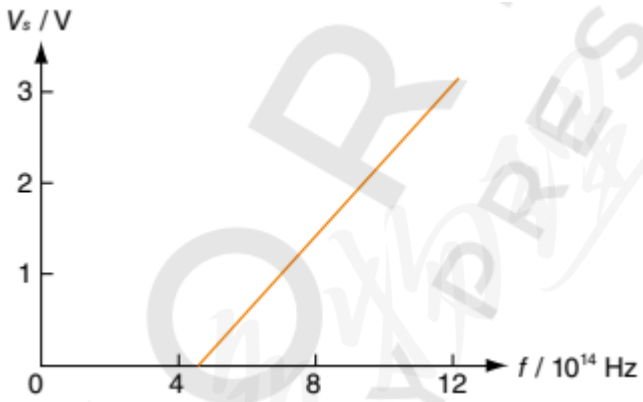
Reverse voltage prevents photoelectrons reaching the collector; at the stopping potential the photocurrent just reaches zero.

E2.1.5 Stopping-potential-frequency graphs

Above threshold, stopping potential varies linearly with frequency for a fixed material. The slope determines Planck's constant and the frequency-axis intercept gives threshold frequency. The extrapolated negative-potential region is mathematical only, not emission below threshold. Changing material changes intercepts, while the ideal-model slope stays unchanged.

$$V_s = \frac{h}{e}f - \frac{\phi}{e}$$

For $f \geq f_0$: V_s is stopping-potential magnitude (V); slope h/e; intercept $-\varphi/e$. $h = 6.63 \times 10^{-34}$ J s; $e = 1.60 \times 10^{-19}$ C; φ: J.



Stopping potential varies linearly with frequency in the region where photoelectric emission occurs.

E2.1.6 Wave-particle duality of light

Interference and diffraction are described by light's wave behaviour, while discrete energy transfers in the photoelectric effect require photons. These descriptions address different experimental features. Choose the model appropriate to the observed phenomenon.